## 11.3 Types of Functions

Functions can be classified into two types,

1. Pre-defined or Built-in or Library Functions
2. User-defined Function.

C++ provides a rich collection of functions ready to be used for various tasks. The tasks to be performed by each of these are already written, debugged and compiled, their definitions alone are grouped and stored in files called **header files**. Such ready-to-use sub programs are called **pre-defined functions or built-in functions.**

C++ also provides the facility to create new functions for specific task as per user requirement. The name of the task and data required (arguments) are decided by the user and hence they are known as **User-defined functions.**

## 11.4 C++ Header Files and Built-in Functions

Header files provide function prototype and definitions for library functions. Data types and constants used with the library functions are also defined in them. A header file can be identified by their file extension **.h.** A single header file may contain multiple built-in functions.

For example: **stdio.h** is a header file that contains pre-defined **"standard input/ output"** functions.

### 11.4.1 Standard input/output (stdio.h)

This header file defines the standard I/O predefined functions **getchar(), putchar(), gets(), puts()** and etc.

#### 11.4.1.1 getchar() and putchar() functions

The predefined function **getchar()** is used to get a single character from keyboard and **putchar()** function is used to display it.

**Program 11.1 C++ code to accept a character and display it**

```
#include<iostream>
#include<stdio.h>
using namespace std;
int main()
{
        cout<<"\n Type a Character : ";
        char ch = getchar();
        cout << "\n The entered Character is: ";
        putchar(ch);
        return 0;
}
```

**Output:**

```
Type a Character : T
The entered Character is: T
```

#### 11.4.1.2. gets() and puts() functions

Function gets() reads a string from standard input and stores it into the string pointed by the variable. Function **puts()** prints the string read by **gets()** function in a newline.



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